

MANDADI VENU MADHAV REDDY

Computer Science Engineering Graduate | Artificial Intelligence and Machine Learning | 2026

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SUMMARY

Computer Science graduate in Artificial Intelligence and Machine Learning, looking for a Java backend or software engineering role. Main project is a multi-agent order processing backend in Spring Boot, built to understand how a retail company handles stock, payment and delivery behind the scenes. Writes tests, documents how each project runs, and keeps all the code on GitHub.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C

Backend and APIs: Spring Boot, Spring MVC, Jakarta Bean Validation, Java records, dependency injection, Maven, multi-agent orchestration, REST API design, FastAPI, Flask, Node.js, Express.js, JWT

Data and Frontend: scikit-learn, pandas, NumPy, model evaluation, DuckDB, Power BI, DAX, Next.js, React

DevOps and Testing: Docker, Docker Compose, GitHub Actions CI, Git, Vercel, JUnit, pytest, Vitest, Postman

PROJECTS

Sampoorna Retail Backend (multi-agent order system, live)

Jul 2026

Java 17, Spring Boot, Maven, Jakarta Bean Validation, Python, FastAPI, Next.js, Docker, GitHub Actions

- Built this to learn how a retail company processes an order behind the scenes, so the work sits in the backend rather than the interface. A Java Spring Boot service runs each order through six agents, with a Python FastAPI service for product suggestions and Next.js exposing the API routes.
- Wrote an orchestrator that owns the whole flow. The agents never call each other; one coordinator calls them in order and decides what happens next, after Bean Validation rejects any invalid order. Catalogue reserves the stock, Risk checks the address and order value, Payment authorises UPI, card or cash on delivery, Fulfilment plans the picking and packing, Delivery builds an eight-stage tracking timeline, and Notification saves the final result.
- Made the agents communicate through one shared contract. Each returns the same Java record with its name, a status of completed, failed or skipped, a message and its duration, plus its own result such as the payment reference or delivery timeline. The orchestrator merges these into one response with a step-by-step trace, so anyone can see which agent did what and where an order stopped.
- Treated stock as a state machine, not a number. Each item records stock before, quantity held and stock after. A confirmed order commits the stock, a declined payment releases it back so the item is available again, and an out-of-stock item is rejected with nothing held, so stock is never lost or double counted. Out-of-stock orders stop before payment is taken.
- Added a chat agent that reads a customer question, decides which of seven actions it needs (search, compare, view cart, checkout, find an order, list orders, help) and runs that action against the real catalogue, cart and order data, so it cannot invent a price or a stock figure.
- Packaged all three services with Docker and Docker Compose, ran JUnit, pytest and Vitest as three parallel GitHub Actions jobs on every push, and kept a fallback path so orders complete if the Java service is down.

AI Source-Code Vulnerability Detection

Apr 2026

Python, scikit-learn, DiverseVul, GitHub Actions

- Trained a model to find security bugs in C and C++ code from a public dataset of 330,492 real functions across 741 open-source projects, scoring 0.335 F1 on 33,000 unseen functions against published research that reaches 0.472.
- Found 1,425 training functions that were also in the test data and removed them before training, since they would have made the score look better than it was. Wrote 92 tests covering the code.

Multi-City Weather Forecasting

Jan 2026

Python, time series, Open-Meteo, scikit-learn

- Prepared 1.75 million hourly weather readings for 20 cities into a daily dataset, trained only on earlier years and tested on 2025 which the model had never seen, giving an average temperature error of 0.97 degrees Celsius.

Retail Sales KPI Analytics

Oct 2025

Python, SQL, DuckDB, Power BI, DAX

- Cleaned 1.07 million sales records, organised them in a DuckDB database using SQL, and built six Power BI report pages covering revenue, monthly trend, product mix, returns and data quality.

EXPERIENCE

Web Development Intern, Prodigy Infotech

Mar 2024 - Apr 2024

Remote

- Built responsive React, JavaScript, HTML and CSS interfaces from written requirements, checked them across desktop and mobile layouts, and delivered two complete working web applications using Git and GitHub.

EDUCATION

B.Tech, Computer Science and Engineering (AI and ML), KLH University | GPA 8.78 / 10

Jul 2022 - Jun 2026

Intermediate (Class 12), Sri Chaitanya Junior College | 92.4%

Jun 2020 - May 2022

SSC (Class 10), Alpha High School | GPA 10.0 / 10

May 2020

CERTIFICATIONS

AWS Certified Cloud Practitioner | MongoDB Python Developer Path | Automation Anywhere RPA Certification